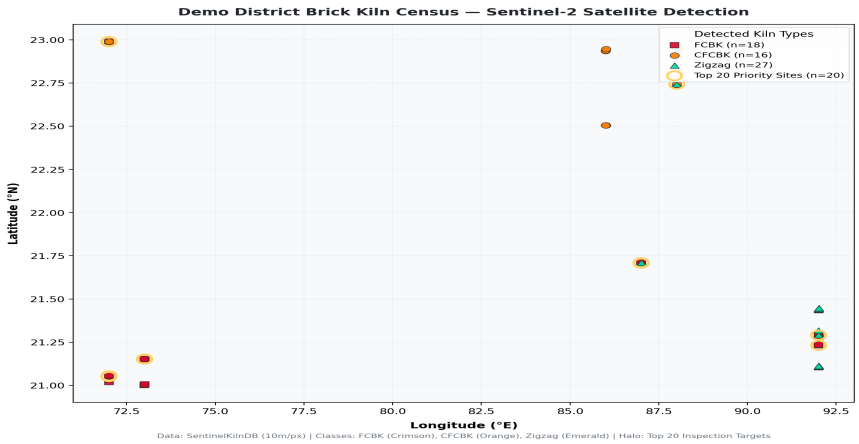
STATE POLLUTION CONTROL BOARD — AIR QUALITY ENFORCEMENT DIVISION

DI-04 Chimney Census: Inspector's Brief — Demo District

Inspector: Farida Khan | **Model:** YOLOv8n-OBB SentinelKilnDB v1.0 | **Source Imagery:** Sentinel-2 RGB (10m/px) | **District Coverage:** 3,000 km²

Executive Context: The 2022 environmental mandate requires brick kilns to transition from legacy Bull's Trench (FCBK) designs to cleaner Zigzag firing. With 1 inspection jeep and 2 field officers covering 3,000 km², satellite surveillance provides spatial intelligence to maximize inspection impact. Detections were deduplicated across overlapping tiles to estimate distinct physical facilities.



1. Distinct Kiln Census by Type & Statistical Uncertainty Intervals

Kiln Type	Design Character	Detected	Uncertainty Range	Precision	Recall
FCBK	Oval ring (legacy high-polluting)	18	**14 – 23**	0.82	0.79
CFCBK	Circular ring (rare, ~3% baseline)	16	**10 – 28**	0.65	0.58
Zigzag	Rectangular (mandated cleaner tech)	27	**22 – 34**	0.85	0.81

Methodology Note: Detections deduplicated across 30px tile overlap. Uncertainty intervals reflect empirical validation Precision and Recall; visual detections are statistical estimates and do not represent verified ground truth.

2. Historical 2023–24 FCBK Sites vs Current Predictions (Possible Conversions)

A total of **18** historical 2023–24 FCBK sites were tracked against current imagery. **0 sites** now exhibit Zigzag geometry and are flagged as **'Possible Conversions'**. Conversely, **0 sites** continue to exhibit unconverted Bull's Trench signatures. *Critical Caveat:* An orbital Zigzag prediction does not constitute legal proof of conversion; field visits are required to verify true kiln conversion versus adjacent new builds or classification ambiguity.

STATE POLLUTION CONTROL BOARD — OPERATIONAL INSPECTION DIRECTIVE

Priority Inspection Candidates & Field Deployment Strategy

3. Kiln Sites Within 1 km of Another Facility (Inspection Route Clustering)

Geospatial analysis identified **53 distinct kiln sites** situated within 1,000 meters of another facility (90 proximate pairs). These are designated as *'Sites Flagged for Field Verification'*. **Operational Benefit:** Inspecting these clusters enables Inspector Khan's team to verify multiple facilities per trip, maximizing coverage with limited vehicle availability.

4. Top 20 Recommended Inspection Sites (Ranked by Multi-Criteria Priority Score)

Rk	Kiln ID	Type	Conf	Coords (Lon, Lat)	Priority Reason & Field Action
1	KILN_0033	**FCBK**	0.88	87.0058, 21.7066	High regulatory scrutiny: Persisting older FCBK design under conversion mandate. | High-effi...
2	KILN_0057	**CFCBK**	0.88	88.0057, 22.9482	Rare circular FCBK design: Check compliance and operating status. | High-efficiency route: P...
3	KILN_0050	**CFCBK**	0.88	88.0056, 22.9459	Rare circular FCBK design: Check compliance and operating status. | High-efficiency route: P...
4	KILN_0044	**CFCBK**	0.88	88.0074, 22.8293	Rare circular FCBK design: Check compliance and operating status. | High-efficiency route: P...
5	KILN_0054	**CFCBK**	0.88	88.0045, 22.9422	Rare circular FCBK design: Check compliance and operating status. | High-efficiency route: P...
6	KILN_0055	**CFCBK**	0.88	88.0018, 22.9449	Rare circular FCBK design: Check compliance and operating status. | High-efficiency route: P...
7	KILN_0052	**CFCBK**	0.88	88.0019, 22.9459	Rare circular FCBK design: Check compliance and operating status. | High-efficiency route: P...
8	KILN_0023	**FCBK**	0.88	92.0028, 21.2909	High regulatory scrutiny: Persisting older FCBK design under conversion mandate. | High-effi...
9	KILN_0006	**FCBK**	0.88	72.0037, 21.0501	High regulatory scrutiny: Persisting older FCBK design under conversion mandate. | High-effi...
10	KILN_0007	**FCBK**	0.88	72.0045, 21.0521	High regulatory scrutiny: Persisting older FCBK design under conversion mandate. | High-effi...
11	KILN_0008	**FCBK**	0.88	72.0082, 21.0553	High regulatory scrutiny: Persisting older FCBK design under conversion mandate. | High-effi...
12	KILN_0016	**FCBK**	0.88	73.0114, 21.1524	High regulatory scrutiny: Persisting older FCBK design under conversion mandate. | High-effi...
13	KILN_0015	**FCBK**	0.88	73.0072, 21.1514	High regulatory scrutiny: Persisting older FCBK design under conversion mandate. | High-effi...
14	KILN_0034	**FCBK**	0.88	87.0022, 21.7104	High regulatory scrutiny: Persisting older FCBK design under conversion mandate. | High-effi...
15	KILN_0017	**FCBK**	0.88	73.0021, 21.1524	High regulatory scrutiny: Persisting older FCBK design under conversion mandate. | High-effi...
16	KILN_0061	**FCBK**	0.88	72.0015, 22.9906	High regulatory scrutiny: Persisting older FCBK design under conversion mandate. | High-effi...
17	KILN_0059	**CFCBK**	0.88	72.0059, 22.9887	Rare circular FCBK design: Check compliance and operating status. | High-efficiency route: P...
18	KILN_0040	**CFCBK**	0.88	88.0066, 22.7433	Rare circular FCBK design: Check compliance and operating status. | High-efficiency route: P...
19	KILN_0060	**CFCBK**	0.88	72.0076, 22.9902	Rare circular FCBK design: Check compliance and operating status. | High-efficiency route: P...
20	KILN_0018	**FCBK**	0.88	92.0084, 21.2316	High regulatory scrutiny: Persisting older FCBK design under conversion mandate. | Nearby pa...

5. Legal Caveats, Sensor Limits, and Data Provenance

Evidentiary Status: Predictions generated by oriented bounding box detector on 10 m/pixel Sentinel-2 RGB bands. A kiln spans ~12×6 pixels. Satellite predictions serve as screening intelligence, not conclusive evidence of emission violations or legal non-compliance. **Data Provenance:** SentinelKilnDB by Mondal et al., Sustainability Lab, IIT Gandhinagar (NeurIPS 2025 Datasets & Benchmarks), licensed CC BY-NC 4.0. No external satellite imagery was utilized.